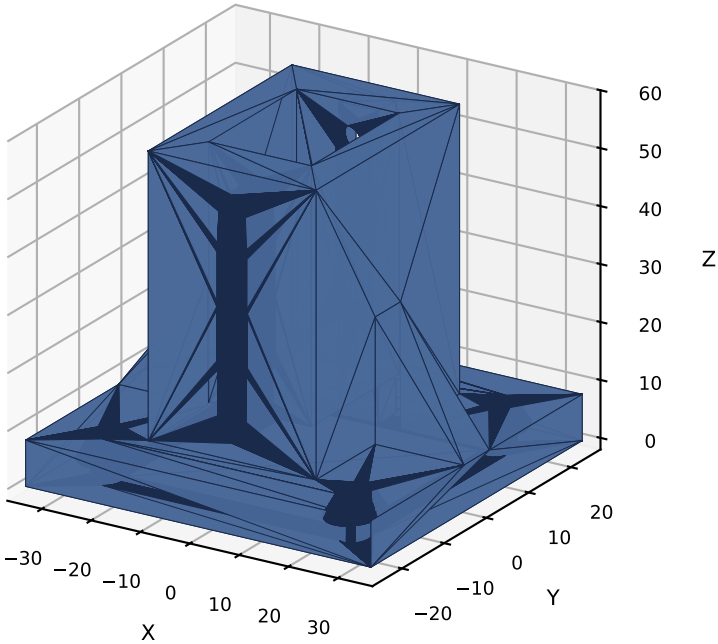


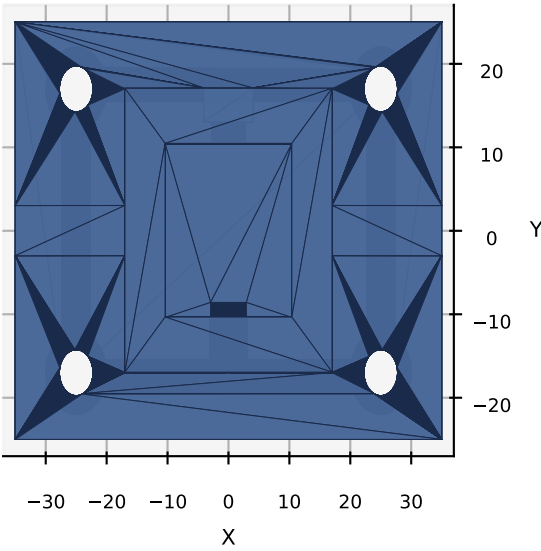
CSM V3 — HMI Module 10 — Mast Base Socket V1.1

Full U-pocket bracket for 2020 V-slot extrusion • Print 2 (left + right) • PETG, 100% infill, 4 walls

Isometric View

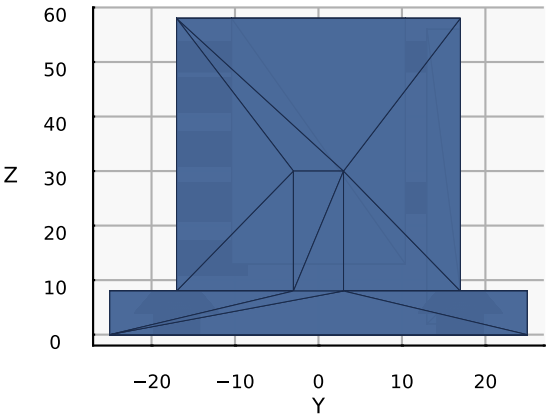
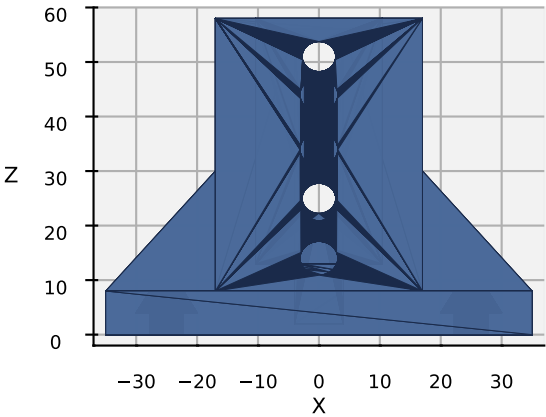


Top View (X-Y plane)



PART	Mast Base Socket
VERSION	V1.1
MODULE	10 — HMI
QTY	2 (left + right)
MATERIAL	PETG (PA12 prod)
INFILL	100%, 4 walls
LAYER	0.2 mm
MASS	~82 g each
TIME	~3-4 h each
INSTALL	(X=±75, Y=-210)
DIMENSIONS (mm)	
Socket outer	34.0 × 34.0 × 50.0
Pocket inner	20.8 × 20.8 (0.4 dia clear)
Wall thickness	6.6
Foot	70.0 × 50.0 × 8.0
Foot bolt grid	±25.0 × ±17.0
Lateral bolt Z	12.0 / 38.0
Gusset (B×H×T)	18.0 × 22.0 × 6
Cable channel	8.0 × 4.0 (rear)
Insert pocket Ø	6.2 × 8.5 deep
Insert count/pitch	4 × 10.0 (front)

Front View (-Y face, operator side) — 4× insert pocketsSide View (+X face) — gusset profile + cable channel



ASSEMBLY (per bracket)

1. Print PETG foot-down, 100% infill, 4 walls, 0.2 mm
2. Heat 4× M5 brass inserts (B0DPQJ4W3Z) into FRONT face pockets with soldering iron at ~220 °C, ~30 s each
3. Slide 2× M5 T-nuts into 2020 FRONT slot at desired Z
4. Slide bracket pocket down over 2020 base
5. M5 × 12 mm bolts through bracket FRONT into T-nuts
6. Place bracket on wood base at (X=+75 OR -75, Y=-210)
7. Mark + drill 4× Ø5.5 in wood under foot bolt positions
8. M5 × 25 socket-head bolts through countersunk foot top
9. Steel flat washer + M5 nyloc nut underneath wood base
10. Verify 2020 stands vertical (square against plate)

EXPANSION (future, via heat-set inserts):

- Camera mount, LED strip, status panel, sensors
- Repeatable install/remove with M5 screws

STRESS NOTES:

- 50 mm socket engagement (V1.0 had 35 mm)
- Lateral bolts spaced 12/38 mm for vertical retention
- Gussets both X sides for foot-socket transition
- Foot-socket fillet attempted but skipped (OCC error)
- Manual blend candidate if first-print shows cracks